MILITARY SPECIFICATION SHEET

POWDER MATERIALS FOR FLAME AND PLASMA SPRAYED COATINGS ALUMINUM SILICON ALLOY, TYPE I, CLASS 2, COMPOSITION E

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the powder described herein shall consist of this document and the latest issue of MIL-P-85856.

Chemical Composition

Fe	Al	Si	Cu	Zn	Mg	Others (1)	
						Each	Total
0.8 max.	Bal	11.0 13.0	0.3 max.	0.2 max.	0.1 max.	0.05 max.	0.15 max.

(1) Determination not required for routine acceptance.

Particle Size/Distribution

+140 Mesh (+106 µm)	1.0 max.
+170 Mesh (+90 µm)	7.0 max.
+325 Mesh (+45 µm)	90.0 min.

Sprayed Coating Properties

Bond Strength	10.3 MPa (1500 psi) min.
Hardness	Rockwell 15T, 60 min.
Microstructure	See paragraph 4.5.4.1.2 Porosity - 15% max. Unmelted Particles - 15% max.

AMSC N/A FSC 3439

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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Deposit Efficiency and Deposition Rate

65% min for N_2/H_2 @ 10.0 lb/hr
60% min for A/H_2 @ 10.0 lb/hr

Comparable Documents

Powder which meets the requirements specified is qualified under the following specifications.

General Electric Company	B50TF92, Class A
Pratt & Whitney Aircraft	PWA 1335
Allison Gas Turbine	EMS 56766

Custodians:
Air Force - 20
Navy - AS
Army - MR

Review activities:
Air Force - 99, 71
Navy - SH

Preparing activity:
Navy - AS
(Project No. 3439-0663/04)

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